

AYUSH KUMAR JHA B

+91-6379722206 | ayushjha3108@gmail.com

github.com/Ak07-cyber | linkedin.com/in/kumar-ayush-jha0007 | ayush-jha-0-portfolio.netlify.app

Full-Stack Developer | AI & LLM Integrations | Cloud & DevOps

SUMMARY

Full-stack developer with production experience shipping **AI-integrated web platforms** across education, healthcare, university-governance, and consumer domains. Strong in **React, Next.js, Node.js, and TypeScript** with hands-on **LLM, RAG, and embedding work (Gemini, OpenAI, LLaMA 3, Sarvam AI)** on **Supabase/PostgreSQL** backends. Comfortable owning features end-to-end – REST APIs, authentication, containerization, and CI/CD – and delivering reliable, ATS-clean, production-ready systems.

EDUCATION

Chennai Institute of Technology

Nov 2022 – Apr 2026

B.E. in Computer Science and Engineering (AI & ML)

CGPA: 8.75 / 10

- Relevant Coursework: Data Structures, Algorithms, DBMS, Operating Systems, Computer Networks, Machine Learning.

EXPERIENCE

Full-Stack Developer

Jan 2025 – Present

DreamPi (a venture of Draxel Product Innovations)

Chennai, TN

- Delivered **production, multi-tenant AI platforms** for education, healthcare, and university-governance clients using **Next.js, Node.js, Supabase, and Google Gemini**, owning features from design through deployment.
- Built **RAG pipelines (pgvector + Gemini/HuggingFace embeddings)** and voice-AI flows (**Sarvam AI** STT/TTS) that power in-app AI tutors and clinical/academic assistants.
- Implemented **role-based access control**, multi-tenant **RLS**, OAuth/phone-OTP auth, and insert-only audit logging across student, teacher, admin, and government portals.
- Integrated third-party services (**Razorpay, DigiLocker, APAAR/ABC**) and set up **GitHub Actions** CI with typecheck and production-build gates on every PR.

AI Engineer Intern

May 2024 – Jul 2024

Axcord Global Pvt. Limited

Bangalore

- Built an **embedding-based semantic search** system backed by a vector database, indexing **120K+** records and tuning it for speed and relevance.
- Layered a **LLaMA 3** response layer to turn retrieval results into concise natural-language recommendations and explanations.
- Refined indexing, metadata filtering, and similarity ranking – improved top-5 relevance from **61% to 72%** and cut average query latency from **320ms to 272ms**.

Front-End Developer Intern

Nov 2023 – Dec 2023

Alam Buhari Enterprise

Chennai, TN

- Built responsive web interfaces in **HTML, CSS, and JavaScript** with consistent cross-browser behaviour across desktop and mobile.
- Optimized UI components and asset loading, raising Lighthouse Performance from **68 to 78 (15%)**.
- Debugged and resolved cross-browser layout and JavaScript issues, improving stability across releases.

PROJECTS

VeluraHQ | Next.js 15, React 19, TypeScript, Supabase, Gemini, MCP

- Key contributor building the authenticated **workspace frontend** of an AI-native SEO/AEO marketing platform using **Next.js 15 (App Router), React 19, TypeScript, Tailwind, and shadcn/ui**.
- Built the node-based **workflow / SuperBlocks engine** (research → refresh → optimize → review gates → CMS publish) with a typed block catalog and graph runner, plus **Power Sheets** for bulk execution across hundreds of URLs.
- Integrated **Pragya**, an in-dashboard AI assistant using **Gemini function calling** over an in-process **MCP server** exposing workspace tools on **Supabase**; powered AI-search tracking of citations, sentiment, and share-of-answer across **ChatGPT, Perplexity, Gemini, Claude, and Google AI Overviews**.

THAAKAM | Next.js, Supabase, Gemini, Sarvam AI, pgvector RAG

- Built a bilingual (English/Tamil) AI education platform aligned with the **Samacheer Kalvi** curriculum, spanning student, teacher, parent, admin, and government portals on a multi-tenant **Supabase** backend (RLS, OAuth).

- Engineered a production **RAG pipeline** (pgvector + Gemini embeddings) with **Sarvam AI** for STT/TTS, OCR, and regional-language reasoning to power AI-assisted learning.

Nadi HMS | *Next.js, Supabase, Google Gemini, MCP, FHIR R4*

- Contributed to a full-stack hospital management platform spanning **60+ clinical/operational modules** (OPD, IPD, billing, lab, pharmacy, ICU, OT) on a Supabase backend.
- Integrated **Asha**, a **Gemini** clinical assistant (chat, voice/STT, discharge drafting, TPA claims), and an **MCP server** exposing hospital APIs as AI-callable tools.

Vidya UMS | *Next.js, Supabase, Google Gemini, MCP*

- Helped build an AI-native multi-tenant University Management System covering admissions, academics, fees, and **NAAC/NBA/NIRF** compliance.
- Wired a **Gemini** AI agent runtime and an MCP-powered admin console integrating **DigiLocker, APAAR, ABC, and Razorpay** under role-based access and audit logging.

AI-Powered Interview Preparation Platform | *React, Node.js, TypeScript, MongoDB, Gemini API* [GitHub](#)

- Built a full-stack platform generating **personalized interview reports** from resumes and job descriptions, cutting prep time by **4+ hours/user**.
- Engineered a **Gemini API** pipeline producing role-fit scores, skill-gap insights, and learning roadmaps from parsed PDF resumes.
- Implemented **JWT + HTTP-only cookie** auth with protected REST APIs (token blacklisting/expiry via Redis); containerized with **Docker Compose** and automated CI with **GitHub Actions**.

BimaSarthi: AI-Powered Insurance Platform | *React, Node.js, TypeScript, OpenAI, AWS ECS Fargate* [GitHub](#)

- Built an AI recommendation engine matching users to insurance plans via income, family profile, and financial goals – cutting decision time by **70%**.
- Engineered an **OpenAI + OCR** pipeline to automate policy analysis and flag hidden clauses; deployed containerized services on **AWS ECS Fargate** with auto-scaling.

Muzer: Collaborative Music Streaming SaaS | *Next.js, Prisma, PostgreSQL, NextAuth, Docker* [GitHub](#)

- Built a multi-tenant SaaS where users create music spaces and queue/vote on songs with real-time, priority-based playback that auto-surfaces the highest-voted track.
- Secured sessions with **NextAuth**, containerized via **Docker**, and shipped through **GitHub Actions** CI/CD to **AWS ECS Fargate** with auto-scaling.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, C/C++, Python, SQL

Frameworks & Libraries: React, Next.js, Node.js, Express.js, FastAPI, Tailwind CSS, WebSockets, WebRTC

AI & ML: LLM APIs (Gemini, OpenAI, LLaMA 3, Sarvam AI), RAG (pgvector), MCP, Embeddings, Vector Search, Prompt Engineering

Databases & Storage: PostgreSQL, MongoDB, Supabase, Redis, Qdrant, Chroma

Cloud & DevOps: AWS (ECS/Fargate, Auto Scaling), GCP, Docker, Docker Compose, Kubernetes, Nginx, GitHub Actions, CI/CD

Practices: REST APIs, Microservices, Monorepo, RAG, Agile

Monitoring: Prometheus, Grafana, New Relic

Tools: Git, GitHub, Postman, Prisma, TurboRepo

ACHIEVEMENTS

Smart India Hackathon 2024 – Finalist: represented the college among top national teams.

Top Performer in Analytical Proficiency – CDC, Chennai Institute of Technology: ranked among top students across quantitative, logical, and verbal reasoning assessments.

Promotion & Social Media Lead – national-level technical symposium: drove digital marketing, lead generation, and strategic communications.

CERTIFICATIONS

AWS Certified DevOps Engineer – Professional (Udemy) | **AWS Cloud & Fundamentals Specialization** (Coursera) | **CCNA: Introduction to Networks** | **Introduction to LLMs** (NPTEL)

SPOKEN LANGUAGES

English (Proficient) | Hindi (Native) | Maithili (Native) | Tamil (Proficient)